Total power radiated by point charge (relativistic)

$$\frac{d\mathcal{E}}{dt} = \frac{\mu_0 q^2}{6\pi c}(\vec{a})^2$$: non-relativistic result from dipole approx

$$d\mathcal{E} = \frac{\mu_0 q^2}{6\pi c} a^2 \, dt$$

Consider now relativistic particle with $\vec{v}, \vec{a}$.

In the instantaneous rest-frame of the particle

$\vec{v} = 0 \quad , \vec{a} \neq 0$

$p^i = (p^0, \vec{p})$

the time component of the 4-momentum $p^i = \left(\frac{\mathcal{E}}{c}, \vec{p}\right)$

$$dp^0 = \frac{\mu_0 q^2 a^2}{6\pi c} \frac{dt}{c} = \frac{\mu_0 q^2}{6\pi c^3} a^2 \, c\,dt$$

We want to express the result in terms of Lorentz-invariant combinations only

Interval $ds = c\,d\tau = \frac{c\,dt}{\gamma} \quad \left(\gamma = \frac{1}{\sqrt{1-v^2/c^2}}\right)$

4-velocity $u^k = \frac{dx^k}{ds} = \left(\frac{c\,dt}{c\,dt/\gamma}, \frac{d\vec{x}}{c\,dt/\gamma}\right) = \gamma\left(1, \frac{\vec{v}}{c}\right)$

4-acceleration $w^k = \frac{du^k}{ds} = \frac{d}{\frac{c}{\gamma}dt}\left(\gamma, \frac{\gamma}{c}\vec{v}\right) = \frac{\gamma}{c}\frac{d}{dt}\left(\gamma, \gamma\frac{\vec{v}}{c}\right)$

$$\frac{d}{dt}\gamma = \frac{d}{dt}\frac{1}{\sqrt{1-\frac{v^2}{c^2}}} = -\frac{-2\,\vec{v}\cdot\dot{\vec{v}}/c^2}{\left(1-\frac{v^2}{c^2}\right)^{3/2}} = \gamma^3\frac{\vec{v}\cdot\dot{\vec{v}}}{c^2}$$

$$\frac{d}{dt}\gamma\vec{v} = \gamma^3\left(\frac{\vec{v}\cdot\dot{\vec{v}}}{c^2}\right)\vec{v} + \gamma\dot{\vec{v}}$$

$$\Rightarrow w^k = \frac{\gamma}{c}\left(\gamma^3\frac{\vec{v}\cdot\dot{\vec{v}}}{c^2}, \gamma\frac{\dot{\vec{v}}}{c} + \gamma^3\frac{\vec{v}\cdot\dot{\vec{v}}}{c^3}\vec{v}\right) =$$

$$= \left(\gamma^4\frac{\vec{v}\cdot\dot{\vec{v}}}{c^3}, \gamma^2\frac{\dot{\vec{v}}}{c^2} + \gamma^4\frac{\vec{v}\cdot\dot{\vec{v}}}{c^4}\vec{v}\right)$$

Can check that $w^k u_k = 0$.

In the rest frame of the charge $\vec{v} = 0$, $\gamma = 1 \Rightarrow$

$$w^k = \left(0, \frac{\vec{a}}{c^2}\right) \qquad \vec{a} = \dot{\vec{v}}$$

$$\Rightarrow \quad w^k w_k = -\frac{a^2}{c^4}$$

$\Rightarrow$ can write $dp^0 = \dfrac{\mu_0 q^2}{6\pi c^3}\left(-w^k w_k c^4\right) \underbrace{c\,dt}_{dx^0}$

$$\Rightarrow dp^i = -\frac{\mu_0 q^2 c}{6\pi} w^k w_k \, dx^i$$

4-vectors

Therefore in the arbitrary frame: $(\dot{\vec{v}} = \vec{a})$

$$w^k w_k = \frac{\gamma^8}{c^6}(\vec{v}\cdot\vec{a})^2 - \left(\frac{\gamma^2}{c^2}\vec{a} + \frac{\gamma^4}{c^4}(\vec{v}\cdot\vec{a})\vec{v}\right)^2$$

$$= \frac{\gamma^8}{c^6}(\vec{v}\cdot\vec{a})^2 - \left(\frac{\gamma^4}{c^4}a^2 + 2\frac{\gamma^6}{c^6}(\vec{v}\cdot\vec{a})^2 + \frac{\gamma^8}{c^8}(\vec{v}\cdot\vec{a})^2 v^2\right)$$

$$= (\vec{v}\cdot\vec{a})^2\left(\frac{\gamma^8}{c^6} - \frac{2\gamma^6}{c^6} - \frac{\gamma^8}{c^6}\frac{v^2}{c^2}\right) - \frac{\gamma^4}{c^4}a^2$$

$$\frac{\gamma^8}{c^6}\left(1 - \frac{v^2}{c^2}\right) = \frac{\gamma^6}{c^6}$$

$$w^k w_k = -\left(\frac{\gamma^4}{c^4}a^2 + \frac{\gamma^6}{c^6}(\vec{v}\cdot\vec{a})^2\right)$$

Observe that $(\vec{v}\times\vec{a})^2 = v^2 a^2 \sin^2\theta = v^2 a^2 (1 - \cos^2\theta)$

$$= v^2 a^2 - (\vec{v}\cdot\vec{a})^2$$

Then $w^k w_k = -\frac{\gamma^6}{c^6}\left(\frac{c^2}{\gamma^2}a^2 + (\vec{v}\cdot\vec{a})^2\right) =$

$$= -\frac{\gamma^6}{c^6}\left(c^2\left(1-\frac{v^2}{c^2}\right)a^2 + (\vec{v}\cdot\vec{a})^2\right) = -\frac{\gamma^6}{c^6}\left(c^2a^2 - v^2a^2 + (\vec{v}\cdot\vec{a})^2\right)$$

$$= -\frac{\gamma^6}{c^6}\left(c^2a^2 - (\vec{v}\times\vec{a})^2\right) = -\frac{\gamma^6}{c^4}\left(a^2 - \frac{(\vec{v}\times\vec{a})^2}{c^2}\right)$$

$$\Rightarrow dp^i = \frac{\mu_0 q^2 c}{6\pi}\frac{\gamma^6}{c^4}\left(a^2 - \frac{(\vec{v}\times\vec{a})^2}{c^2}\right)dx^i$$

$$dp^0 = d\left(\frac{\varepsilon}{c}\right) = \frac{\mu_0 q^2 c}{6\pi}\frac{\gamma^6}{c^4}\left(a^2 - \frac{(\vec{v}\times\vec{a})^2}{c^2}\right)\underbrace{c\,dt}_{dx^0}$$

$$d\varepsilon = \frac{\mu_0 q^2}{6\pi c}\left(a^2 - \left(\frac{\vec{v}\times\vec{a}}{c}\right)^2\right)dt$$

Since $\vec{p} = \frac{\vec{v}}{c^2}\varepsilon$

$$d\vec{p} = \frac{\vec{v}}{c^2}d\varepsilon$$

And finally, since $dx^i = u^i\,ds$,

$$dp^i = \frac{\mu_0 q^2 \gamma^6}{6\pi c^3}\left[a^2 - \left(\frac{\vec{v}\times\vec{a}}{c}\right)^2\right]u^i\,ds$$

or most generally

$$dp^i = -\frac{\mu_0 q^2 c}{6\pi}w^k w_k\,u^i\,ds = -\frac{\mu_0 q^2 c}{6\pi}\frac{du^k}{ds}\frac{du_k}{ds}u^i\,ds$$

Use relativistic equ of motion

$$mc \frac{du^k}{ds} = q F^{kl} u_l$$

$$F^{\alpha\beta} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

$$ds = c\,d\tau = \frac{c}{\gamma} dt \Rightarrow mc \frac{d}{ds} = mc \frac{\gamma}{c} \frac{d}{dt}$$

$$m\gamma \frac{du^k}{dt} = q F^{kl} u_l \Rightarrow \begin{cases} m\gamma \dfrac{d\gamma}{dt} = \gamma q \dfrac{\vec{v}\cdot\vec{E}}{c^2}, & k=0 \\ m\gamma \dfrac{d\vec{p}/c}{dt} = \dfrac{\gamma}{c} q \left(\vec{E} + \vec{v}\times\vec{B}\right), & k=1,2,3 \end{cases}$$

$$dp^i = -\frac{\mu_0 q^2 c}{6\pi} \left(\frac{q}{mc} F^{kl} u_l\right)\left(\frac{q}{mc} F_{km} u^m\right) dx^i$$

Note that $F^{kl} u_l = f^k = (f^0, +\vec{f})$

$$F_{kn} u^n = f_k = (f^0, -\vec{f})$$

$$\Rightarrow \left(F^{kl} u_l\right)\left(F_{km} u^m\right) = (f^0)^2 - (\vec{f})^2$$

where $f^0 = \dfrac{\gamma}{c^2} \vec{v}\cdot\vec{E}$

$$\vec{f} = \gamma \frac{\left(\vec{E} + \vec{v}\times\vec{B}\right)}{c}$$

$$dp^i = \frac{\mu_0 q^2 c}{6\pi}\left(\frac{q}{mc}\right)^2 \left\{ \frac{\gamma}{c}(\vec{E}+\vec{v}\times\vec{B}) \cdot \frac{\gamma}{c}(\vec{E}+\vec{v}\times\vec{B}) - \left(\frac{\gamma}{c}\,\frac{\vec{v}\cdot\vec{E}}{c}\right)^2 \right\} dx^i$$

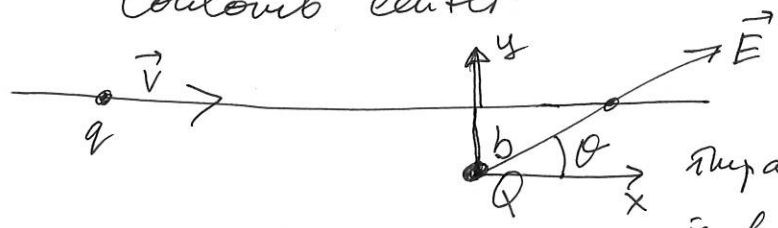
$$dp^i = \frac{\mu_0 q^4}{6\pi m^2 c^3}\,\gamma^2 \left\{ (\vec{E}+\vec{v}\times\vec{B})^2 - (\vec{v}\cdot\vec{E}/c)^2 \right\} dx^i$$

$$\Rightarrow \quad dp^0 = d\frac{\mathcal{E}}{c} = \frac{\mu_0 q^4}{6\pi m^2 c^3}\,\frac{(\vec{E}+\vec{v}\times\vec{B})^2 - \left(\frac{\vec{v}\cdot\vec{E}}{c}\right)^2}{1-v^2/c^2}\, c\,dt$$

$$\Delta\mathcal{E} = \frac{\mu_0 q^4}{6\pi m^2 c}\int_{-\infty}^{\infty} dt\,\frac{(\vec{E}+\vec{v}\times\vec{B})^2 - \frac{1}{c^2}(\vec{v}\cdot\vec{E})^2}{1-v^2/c^2}$$

EM energy radiated by charged particle moving through fields $\vec{E}$, $\vec{B}$.

Example: relativistic particle scattering off Coulomb center



impact parameter b is large

Q creates $\vec{E} = \dfrac{Q\,\vec{r}}{4\pi\varepsilon_0 r^3}$

Consider $v \approx c$ so that trajectory of the particle q is straight line $\vec{x} = vt\,\hat{x}$ $\quad (\vec{v} = v\,\hat{x})$

$\Rightarrow r = \sqrt{b^2 + v^2 t^2}$

$$\Delta \mathcal{E} = \frac{\mu_0 q^4}{6\pi m^2 c} \int_{-\infty}^{\infty} dt \; \frac{E^2 - \frac{1}{c^2}(\vec{v}\cdot\vec{E})^2}{1 - v^2/c^2}$$

$\vec{v}\cdot\vec{E} = v\,\hat{x}\cdot\vec{E} = v E \cos\theta$, but $\cos\theta = \dfrac{vt}{\sqrt{b^2+v^2t^2}}$

$$\Delta \mathcal{E} = \frac{\mu_0 q^4}{6\pi m^2 c}\left(\frac{Q}{4\pi\varepsilon_0}\right)^2 \gamma^2 \int_{-\infty}^{\infty} dt \left[\left(\frac{1}{r^2}\right)^2 - \frac{1}{c^2}\left(\frac{v\cos\theta}{r^2}\right)^2\right]$$

$$= \frac{\mu_0 q^4}{6\pi m^2 c}\left(\frac{Q}{4\pi\varepsilon_0}\right)^2 \frac{\gamma^2}{v} \int_{-\infty}^{\infty} d(vt) \left\{ \frac{1}{(b^2+v^2t^2)^2} - \right.$$

$$\left. - \frac{v^2}{c^2}\frac{(vt)^2}{(b^2+v^2t^2)^3} \right\}$$

$$\int_0^\infty dx \; \frac{1}{(b^2+x^2)^2} = \frac{\pi}{4b^3}$$

$$\int_0^\infty dx \; \frac{x^2}{(b^2+x^2)^3} = \frac{\pi}{16b^3}$$

$$\Rightarrow \quad \Delta \mathcal{E} = \frac{\mu_0 q^4}{6\pi m^2 c} \left(\frac{Q}{4\pi\varepsilon_0}\right)^2 \frac{\gamma^2}{v} \; 2 \cdot \frac{\pi}{4b^3}\left(1 - \frac{v^2}{4c^2}\right)$$

$$= \frac{\mu_0 q^4}{6\pi m^2 c} \left(\frac{Q}{4\pi\varepsilon_0}\right)^2 \frac{\pi}{2b^3 v} \; \frac{4c^2 - v^2}{c^2 - v^2}$$